Design and Development of an Automated Blind Signal Receiver

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1 Outline:

This project outlines the development of an intelligent, "Blind" Signal receiver. The goal is to create a system that can intercept unknown digital radio signals, identify their properties using Deep Learning, and extract the underlying data without having any prior knowledge of the transmitter's settings. The project transitions from simple narrowband signals to complex broadband signals and concludes with real-world hardware implementation.

Part A: Single Carrier Analysis and Blind Reception

The focus here is on narrow-band signals ($\leq$ 300 kHz) with no prior information.

1.1 Objective

The first phase focuses on "Narrowband" signals (signals with a bandwidth of less than 300 KHz). These are typically used for voice or low-speed data transmission

1.2 Deep Learning Modulation Classification

This system uses Deep Learning (Neural Networks) to automatically recognize:

1. **m-PSK (Phase Shift Keying)**: Data mapped to wave phase changes.
2. **n-FSK (Frequency Shift Keying)**: Data mapped to frequency shifts.
3. **p-QAM (Quadrature Amplitude Modulation)**: High-speed data using both amplitude and phase

1.3 Parameter Estimation and Synchronization

Once the signal type is identified, the system must perform "Fine Tuning":

1. **Carrier Synchronization:** Aligning the receiver's frequency to the transmitter's frequency.
2. **Symbol Rate Estimation:** Determining the speed at which the data bits are being sent.
3. **Timing and Phase Correction:** Correcting small errors caused by distance or hardware imperfections.

1.4 Output

The system generates a demodulated bitstream and provides a Confidence Score, which tells the user how reliable the decoded data is.

Part B: Analysis of Complex Broadband OFDM Signals

2.1 Scaling Up

The following part upgrades the system to handle modern, high-speed signals. These have much wider bandwidths and move data at significantly higher speeds (similar to Wi-Fi or LTE).

2.2 OFDM and MIMO Processing

The system transitions from single voices to OFDM (Orthogonal Frequency Division Multiplexing).

1. **OFDM**: Sending data across hundreds of tiny "sub-carriers" simultaneously.
2. **MIMO (Multiple Input Multiple Output):** Utilizing multiple antennas to transmit and receive more data.

2.3 Identification of Communication Standards

The system should distinguish between a Wi-Fi signal, a cellular signal, or a specialized military standard.

2.4 MAC Layer Decoding

The project goes deeper by decoding the MAC Layer. This reveals identifying information like:

1. Sender/Receiver addresses
2. Data packet structures.
3. Network headers.

Part C: System Implementation and Real-World Hardware

3.1 Hardware Integration (USRP SDR)

USRP (Universal Software Radio Peripheral): The specialized radio hardware that connects the antenna to the computer.

3.2 High-Performance Computing

Processing high-speed signals requires immense computer power. The implementation must use Multi-core CPU processing to ensure the algorithms run in "real-time," preventing any lag or loss of incoming data.

3.3 Final Deliverables

The project concludes with the delivery of the full Source Code, demonstrating the ability to turn a standard computer and a USRP antenna into an intelligent, autonomous radio receiver.

2 Implementation

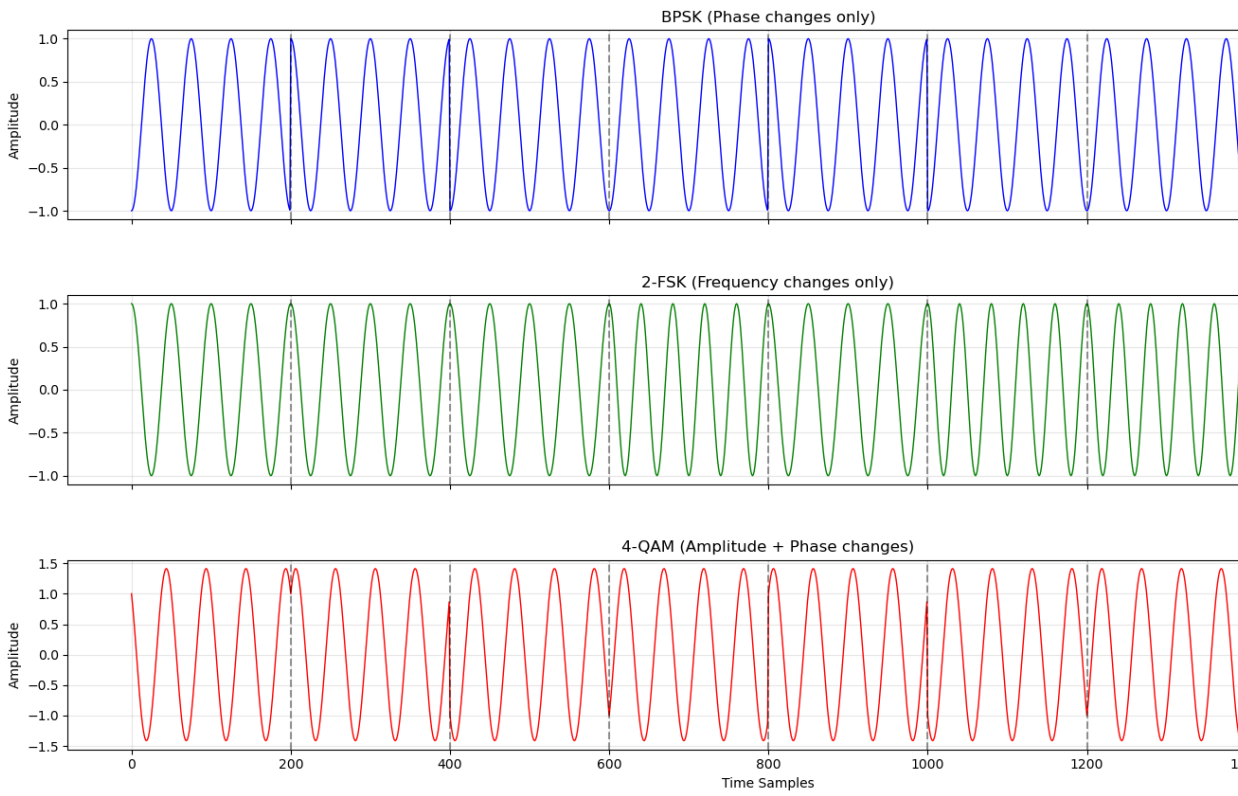

Figure 1: An overview

Note: "4-QAM has constant amplitude but different phase states. Higher-order QAM introduces amplitude variation."

Bit Error Rate (BER)

The Bit Error Rate (BER) is the most fundamental performance metric in digital communications. It gives the probability that a received bit is decoded incorrectly due to noise, interference, or channel impairments.

3 Analysis

3.0.1 Basic Radio Wave

$$s(t) = A\cos(2\pi ft + \phi)$$

Where:

A: Amplitude (Volume / Size)
f: Frequency (Pitch / Speed)
ϕ: Phase (Starting point/offset)

3.0.2 PSK (Phase Shift Keying) Formula

In PSK, we keep the Amplitude (A) and frequency (f) constant. We change the ϕ (Phase).

$$\phi_i = \frac{2\pi(i-1)}{m}, \quad i = 1, 2, ..., m$$

The signal sent is:

$$s_{PSK}(t) = \cos(2\pi f_c t + \phi_i)$$

3.0.3 FSK (Frequency Shift Keying) Formula

The signal sent is:

$$s_{FSK}(t) = \cos(2\pi f_i t)$$

3.0.4 QAM (Quadrature Amplitude Modulation) Formula

QAM combines phase modulation and amplitude modulation. In-Phase (I) and Quadrature (Q) representing two orthogonal carrier signals (i.e cosine and a sine wave) that are out of phase

$$s_{QAM}(t) = I\cos(2\pi f_c t) - Q\sin(2\pi f_c t)$$

Where:
I **(In-phase)**: The value on the X-axis of the constellation grid.
Q **(Quadrature)**: The value on the Y-axis of the constellation grid.
Constellation Grid: A Constellation grid is a 2D map that shows all possible combinations of I and Q values for a specific modulation, such as 16-QAM or 64-QAM which helps us to know health of a transmission.

4 FSK Modulation

1. FSK is a digital modulation technique that converts digital signals into Analog form by varying the carrier's frequency, enabling efficient transmission over a communication channel.
2. In FSK, the Frequency of the Carrier Signal varies according to the Digital Message Signal.

- **For BFSK**

  $$L = 2 = 2^1 = 2^n \rightsquigarrow n = 1 \rightsquigarrow f_{c0}, f_{c1}$$

- **For 4-FSK**

  $$L = 4 = 2^2 = 2^n \rightsquigarrow n = 2 \rightsquigarrow f_{c0}, f_{c1}, f_{c2}, f_{c3}$$

- **For 8-FSK**

  $$L = 8 = 2^3 = 2^n \rightsquigarrow n = 3 \rightsquigarrow f_{c0}, f_{c1}, \ldots, f_{c7}$$

General Formula

$$s_{FSK}(t) = A\cos(2\pi f_i t) \tag{1}$$

$$s_{FSK}(t) = \begin{cases} A\cos(2\pi f_0 t) & \text{bit} = 0 \\ A\cos(2\pi f_1 t) & \text{bit} = 1 \end{cases} \tag{2}$$

4.1 BER

$$\boxed{P_e^{\text{FSK}} = \frac{1}{2}\exp\left(-\frac{E_b}{2N_0}\right)} \tag{3}$$

4.2 Advantages

1. Simple Implementation (Easy Modulation and Demodulation)
2. Better Noise Immunity compared to ASK
3. Reliable for low Data Rate Applications

4.3 Disadvantages

1. Required Large Bandwidth compared to ASK and PSK
2. Coherent Detection more complex
3. Not good for high Data Rate Transmission

4.4 Applications of FSK

- In Early Modems
- Low Frequency Wireless Systems
- Bluetooth (using GFSK variant)

5 BPSK Modulation

Binary Phase Shift Keying (BPSK) is the simplest form of Phase Shift Keying (PSK). It encodes data by shifting the phase of a carrier sinusoid between two states separated by 180°.

5.0.1 Modulated Signal

The general BPSK signal is:

$$s(t) = A\cos\big(2\pi f_c t + \phi(t)\big), \quad 0 \le t \le T_b \tag{4}$$

where

$$\phi(t) = 0 \quad \text{for bit } 1 \tag{5}$$

$$\phi(t) = \pi \quad \text{for bit } 0 \tag{6}$$

This simplifies to:

$$s(t) = \begin{cases} +A\cos(2\pi f_c t) & \text{bit } = 1 \\ -A\cos(2\pi f_c t) & \text{bit } = 0 \end{cases} \tag{7}$$

BER

$$\boxed{P_e^{\text{BPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = \frac{1}{2}\,\text{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)} \tag{8}$$

BPSK gives the **best BER** among all binary schemes at a given E_b/N_0.

5.1 Advantages

- Simple implementation — only two phase states required.
- Robust against amplitude noise and distortion.
- Widely used standard in satellite, deep-space, and wireless communications.

5.2 Disadvantages

- Low spectral efficiency — only 1 bit per symbol.
- Requires coherent detection (precise carrier phase synchronization).
- Not suitable for high-data-rate applications compared to QPSK or QAM.

6 4-QAM Modulation

1. QAM uses four distinct symbols. Each symbol represents 2 bits.
2. QAM has 4 symbols, and each symbol is represented by a distinct phase: $0°, 90°, 180°$, or $270°$.

The 4-QAM transmitted signal is:

$$s_i(t) = A_I \cos(2\pi f_c t) - A_Q \sin(2\pi f_c t), \quad 0 \leq t \leq T_s \tag{9}$$

where $A_I, A_Q \in \{+A, -A\}$ are the in-phase and quadrature amplitudes, f_c is the carrier frequency, and $T_s = 2T_b$ is the symbol duration (each symbol carries 2 bits).

6.1 Constellation Points

The four constellation points are:

$$\mathbf{s} \in \{(+A, +A),\ (+A, -A),\ (-A, +A),\ (-A, -A)\} \tag{10}$$

If Carrier Signal is $C(t) = A\cos(\omega_c t)$, then QAM Signal is given by:

$$\Phi_{\text{QAM}} = \begin{cases} A\cos(\omega_c t + 0°) & \longrightarrow \text{For Symbol 1} = \text{Digital Data 00} \\ A\cos(\omega_c t + 90°) & \longrightarrow \text{For Symbol 2} = \text{Digital Data 10} \\ A\cos(\omega_c t + 180°) & \longrightarrow \text{For Symbol 3} = \text{Digital Data 01} \\ A\cos(\omega_c t + 270°) & \longrightarrow \text{For Symbol 4} = \text{Digital Data 11} \end{cases}$$

1. **Higher Spectral Efficiency than BPSK**
   4-QAM transmits **2 bits per symbol**, doubling the data throughput for the same bandwidth compared to BPSK. The spectral efficiency is:

   $$\eta = \log_2 M = \log_2 4 = 2 \text{ bits/s/Hz} \tag{11}$$

2. **Widely Standardized**
   4-QAM/QPSK is a core modulation in IEEE 802.11 (Wi-Fi), 3GPP LTE/5G, DVB-S2, and Bluetooth LE, ensuring abundant hardware and software support.

6.2 BER

$$\boxed{P_e^{\text{4-QAM}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = P_e^{\text{BPSK}}} \tag{12}$$

6.3 Disadvantages of 4-QAM

1. Compared to 16-QAM (4 bits/symbol) and 64-QAM (6 bits/symbol), 4-QAM is spectrally inefficient when bandwidth is limited.
2. Compared to BPSK, 4-QAM requires an additional quadrature channel, phase splitters, and more complex synchronization circuits, slightly increasing hardware cost and power consumption.

7 Comparison Table

E_b/N_0 **(dB)**	**FSK**	**BPSK**	**4-QAM**
0	3.03×10^{-1}	7.87×10^{-2}	7.87×10^{-2}
2	2.35×10^{-1}	3.75×10^{-2}	3.75×10^{-2}
4	1.65×10^{-1}	1.25×10^{-2}	1.25×10^{-2}
6	1.01×10^{-1}	2.39×10^{-3}	2.39×10^{-3}
8	5.50×10^{-2}	1.91×10^{-4}	1.91×10^{-4}
10	2.05×10^{-2}	3.87×10^{-6}	3.87×10^{-6}
12	5.52×10^{-3}	1.47×10^{-8}	1.47×10^{-8}

1. BPSK and 4-QAM have **identical BER** performance.
2. 4-QAM doubles the throughput of BPSK at **no extra BER cost**.
3. FSK has the **worst BER** but is the simplest to implement as it needs no phase synchronization.